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How Does the Paradoxical Leadership of Cross-Border e-Commerce (CBEC) Gig Workers Influence Chinese Company Performance: The Role of Psychological Well-Being

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Abstract: Cross-border e-commerce (CBEC) has attracted interest in the global marketplace. Meanwhile, CBEC gig workers have changed CBEC enterprises' business models, eliminated outdated enterprises, and reset organizational structures. Consequently, CBEC enterprises must adopt new approaches to promote company performance (CP). However, in Chinese CBEC companies, the psychological well-being (PW) of gig workers is largely ignored. Although some enterprises have paid attention to this problem, the leadership is often not in line with the psychological reality of gig work. Moreover, few studies have focused on the impact of paradoxical leadership (PL) on gig workers' PW in the CBEC industry. Therefore, this study investigates the link between gig workers' PW and organizational performance under PL based on survey data from CBEC enterprises in China. Data were collected from a survey of 346 employees of CBEC enterprises in China. Results showed that PL exerts a significant positive impact on CP. The PW of Chinese CBEC gig workers positively mediates the relationship between PL and CP. Under different dynamic environmental conditions, CBEC enterprises in China should adopt appropriate leadership methods to improve their performance. This study provides new theoretical and practical perspectives for understanding the management of gigs in Chinese CBEC companies.

Keywords: paradoxical leadership; cross-border e-commerce (CBEC); gig workers; company performance; psychological well-being



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1. Introduction

The COVID-19 pandemic has imposed crucial challenges on the normal operation of organizations worldwide. Given the frequently disrupted working schedules and increasing downsizing and pay cuts, temporary and gig workers have become a new trend, particularly in the cross-border e-commerce (CBEC) industry [1], where gig employment is a prevalent form of work arrangement.

Gig workers refer to lowly paid, temporarily contracted employees who generally lack formal training to update their knowledge and skills, as well as social and endowment insurance to ensure a comfortable retirement [2]. Without employers' support to build formal employee–employer relationships, gig workers commonly deal with a toxic workplace environment, which involves cruel and often violent treatment of employees that jeopardizes their safety and health [3]. Thus, we argue that gig workers who constantly suffer from unemployment and are deprived of training and promotion opportunities are especially prone to experiencing serious psychological pressures in the post-pandemic period riddled with uncertainties. Of all the ways to motivate employees, gigs may welcome a more relaxed and freer style of leadership. CBEC is a new industry with a high employee turnover rate. It is greatly influenced not only by the international environment but also by the service quality of its employees. In the case of external environmental uncertainty and

internal staff management instability, the management style of CBEC managers has put forward higher requirements.

Although leadership styles have a wide variety, the current research focuses on the effects of paradoxical leadership (PL) on employee psychological well-being (PW) [4,5]. This is because PL has been found to exert an impact on well-being and performance [6]. As such, it is meaningful to delve into gig workers whose career paths can be described as new, contemporary, and self-directed. As suggested by protean career theory [7], such a modern path of career development is driven by the person according to his/her own goals and psychological success and not by the organization.

The existing literature on the function of PL mainly considers the strategy of resolving organizational tension through continuous learning. At the micro level, the literature focuses on the decentralization and authorization of subordinates. However, there is still an important gap with regard to gig job performance. Although prior literature has considered organizational paradoxes [4], few researchers have attempted to characterize the paradoxical nature of leader behavior in the context of gig management. Moreover, previous research has revealed that happy workers are more productive than unhappy ones, but little is known about the relationship between PL and the PW of gig workers.

The current study aims to fill such gaps by investigating the mechanisms of influence of PL on the PW of gig workers and firm performance in the CBEC industry. Our overall goal is to determine how PL can surpass all existing management models in the CBEC industry and become a more effective way to manage gigs.

(1) Since the 1980s, PL has addressed strategic issues at the macro level [5]. Meanwhile, this study identifies a new direction for PL. We believe that PL should also pay attention to employees' PW at the micro level to proactively address employees' mental health needs and challenges.

(2) Twentieth century management styles, transactional leadership and superior leadership are effective in relatively stable and simple environments but not in complex and dynamic environments, especially those involving gig workers [4,5] and thus require new leadership styles. Structures and needs are often at odds with employees' needs. The requirements are relatively simple and easy to manage in linear and static situations. However, when the environment changes significantly, PL becomes more effective as it can be employed in a complex environment. In sum, this study broadens our perspective on leadership styles in complex environments. PL can generally blur the boundaries of contradictions and accepts holistic thinking. Thus, broadening the cognitive base of "both" is helpful.

(3) On the basis of the reality of gig work in CBEC enterprises in China, this study proposes a new idea of improving the PW of gig workers through PL to directly and effectively promote gig organizational performance [6]. The gig phenomenon has long been neglected in the literature. Few scholars have explored the mental health and PW of gig workers in CBEC enterprises, and the methods for managing this independent group of people have rarely been empirically studied.

(4) The current management of China's CBEC mainly neglects poverty eradication and promotes mutual tolerance and cooperation. This is the factor that affects the sustainable and healthy development of this industry. Changing to a leadership model that is more flexible and inclusive could improve the gig's quality of life because of PL emphasis on the economy and the quality of people's lives and the quality of the environment; social development and ensuring human rights protect culture and respect cultural diversity, ethnicity and religion; and the balance between human and nature, social and human capital and income, and living standards. Such a change is a useful exploration for the health of the industry.

In the current study, we investigate the PW and job performance of gig workers and the improvement of their job performance under PL. This knowledge will not only help managers improve organizational performance in uncertain times but also reveal how gigs

can be made more satisfactory. We study the mechanism of PW under PL and consider the subjective well-being of gig work as the moderating variable.

2. Literature Review and Hypothesis Development

2.1. Protean Career Theory

Protean career theory [7] emphasizes personal motivation in career development, wherein personal needs are more important than organizational needs; hence, it implies that individuals do not need to disregard their goals in order to be loyal to the organization. Take for example individuals who lead the development of organizations but also want to start their own businesses. People in traditional career stages can be redesigned based on this. The theory also emphasizes that all careers should be unique. It is only by developing oneself in one's own field that the greatest professional satisfaction can be obtained, and obstacles in development can be overcome to achieve the highest level. However, the knowledge of one's own talent is a process, and knowledge may change over time. Consequently, everyone's careers may have different peaks, troughs, entrances, and exits. Nevertheless, personal development and lifelong learning are at the core of career development, which makes self-learning increasingly important. Furthermore, individuals can move between different products, technological areas, functions, and organizations. This flow may open many new possibilities. Finally, occupational and biological ages are the most important factors. As protean workers tend to control their own career development to achieve their independent goals and values, they rarely accept external standards or allow others to interfere excessively in their careers [8]. It should be noted that the difference between protean career and gig workers lies in the fact that gig workers lack complete career planning and any guarantees for their own career development and occupational safety. They can hardly rely on themselves to achieve the development of their own careers. They can only be passively chosen by a company, enterprise, industry, or specific work content [8].

2.2. Gig Workers

The origins of "gig" work date back to the pre-industrial era. In 18th century Europe, different systems of trade unions defined the standards for different industries, such as textiles, clothing, footwear, tableware, small furniture, and other simple consumer goods. Businesses delegated production tasks to employees who worked in their homes using their own equipment. The output of their labor was owned by the merchant capitalists who provided the original raw materials and supplies; the producers merely worked on the added value of these materials. These independent workers were paid on a piecework basis and were not guaranteed to perform the same work or work the same hours per week [9,10]. Over time, the content of gig jobs has changed significantly. Journalist Tina Brown first coined the term "gig economy" in 2009. She argues that gig workers pursue a patchwork of freelance projects, consulting and part-time work, and trade in digital marketplaces [11]. Previous researchers have divided gig workers into three categories: app workers, crowd workers, and capital platform workers [12]. In this study, we focus on gig workers engaged in crowd work. Employers select candidates through a platform and decide whether to hire them on the basis of their qualifications, abilities, professional level, and expertise. Gig workers best suited for crowd work are content creators, mobile and web application developers, customer service and sales agents, etc. [11]. These roles are the most important in CBEC. Through cooperation with CBEC enterprises, gig workers may establish partnerships, communicate freely, obtain fixed price protection, and receive hourly payment protection.

However, gig jobs are not always easy. Under algorithmic management, workers are subjected to massive data collection, opaque automated processes, and asymmetric information. In addition, what gig economy platforms say about providing flexibility for employees is often controlled by misleading information and algorithms [13]. However, in reality, workers may face job insecurity, overtime work, irregular sleep schedules, sleep

deprivation, social isolation, and regular behavior and performance tracking. The lack of support from superiors leads to anxiety and confusion, which is the main reason why gig workers leave their jobs. The autonomy that a company prides itself on often becomes a major cause of depression and confusion among employees. In theory, gig workers, as independent contractors, should be able to decide whether to accept assignments [14]. However, owing to strict management, gig workers are often directed and manipulated to take on work that could last beyond their own working hours [15]. Therefore, the psychological conditions of gig workers in CBEC enterprises need more attention as they are closely related to the development of the CBEC industry.

The gig economy business model (also known as the sharing or on-demand economy model) underpinned by artificial intelligence and digital technologies has signified the emergence of various nonconventional forms of work arrangements among stakeholders through online platforms; these arrangements include peer-to-peer nonmonetized sharing and large-scale commercial exchanges (e.g., Uber and Airbnb) [16,17]. Employees under a gig economy paradigm are believed to enjoy far more freedom and flexibility at work and can thus take greater control over their work–life arrangement and balance [18]. This flexible working mode with a high degree of autonomy seems to cater particularly to the tastes of younger people. From a positive perspective, the increasing popularity of this timeless gig work embodies the rise of a new, more entrepreneurial generation with the ambition to manage their career development. However, on the negative side, a sharing economy recruitment model generally offers no promise for long-term future employment, thereby causing job-related concerns pertaining to the function of human resource management (HRM), especially regarding the deprivation of career security among workers [14]. From this perspective, the implementation of gig employment that largely shifts the burden of economic fluctuation and financial risks from employers to employees [19] may be detrimental to the PW of workers.

2.3. Paradoxical Leadership

PL is defined as contradictory but interrelated leadership behaviors that meet competing workplace demands simultaneously [20]. Under PL, leaders require a combination of egocentrism with other-centrism while maintaining distance and intimacy. Moreover, under PL, leaders implement job requirements while promoting flexibility, allowing autonomous colleagues to maintain decision-making control [21], and treating subordinates in a fair but personalized way. In CBEC organizations, managers are expected to push their subordinates and maintain consistent company performance (CP), whereas gig workers expect their superiors to respect their uniqueness. Meanwhile, organizations want managers to control workflow and production processes, whereas gig workers want autonomy and task choice. [22] These challenges can affect managers' effectiveness. Among the many leadership styles, transactional leadership attaches importance to the control of behavior while authoritarian leadership emphasizes decision-making management. Other leadership models emphasize delegation, allow autonomy, and avoid micromanagement. By contrast, PL seeks to reconcile internal tensions [23]. Paradoxical leaders serve as role models to show employees how to accept and embrace complex and contradictory situations [24]. By further exploring the ways to integrate competitive structures and the needs of subordinates, leaders demonstrate, continually learn, and adopt new perspectives to solve seemingly incompatible problems and seek to continuously change the status quo [25]. They also demonstrate to subordinates how to flexibly adapt traditional methods, even the radically unconventional ones, to get people used to changes and encourage them to take the initiative to solve problems. In doing so, subordinates also enhance their ability by constantly responding to possible challenges and expanding their horizons for new opportunities required by the job [24]. In general, PL demonstrates subordinates' ability to learn and become flexible, skilled, adaptable, proactive, and open to work [20].

2.4. CBEC

Some scholars [26] understand e-commerce from a broad perspective, considering it as an international business activity that includes the various impacts of information and communication technologies on enterprises. It includes offering product catalogs, online promotions, online purchases, online payments, online sales services, e-procurement, online marketplaces, and e-fulfillment. Other scholars [27] have taken a narrower view, arguing that e-commerce includes only online sales. E-commerce refers to the process of selling and buying goods or services through commissions. Goods and services are ordered online, but the final delivery of goods or services does not occur online. On the basis of the definition of e-commerce in the United Nations Conference on Trade and Development guide, [28] we consider CBEC as a sales method that uses the internet to sell goods and services to foreign users. CBEC organizations digitize scattered conventional business models, shift and reconfigure organizational structures, and increase employee uncertainties and managerial difficulties [29,30]. An in-depth understanding of the management strategy of coordinating the digital characteristics of knowledge-intensive gig workers with the integration of individual resources needs to be considered. Moreover, CBEC enterprises must change leadership to acclimatize to an unstable and fast-changing environment [31]. Regardless of the business environment, work relationships, and employees, paradoxical leaders tend to accept contradictions as a natural part of life, appreciate their interweaving with nature, and find comfort from the discomfort they cause [32]. Through contradictions, these leaders proactively deal with tension while searching for alternatives. This makes paradoxical leaders more adaptable to the CBEC's environment.

2.5. Employees Psychological Well-Being

Employees' PW is defined as a sign of health and is considered a process of the continuous growth of happiness in the course of life [33]. It is characterized by the presence of positive emotions, absence of negative emotions, and co-existence of job and life satisfaction [34]. Following this, researchers categorize PW into seven dimensions: self-acceptance, personal growth, purpose in life, positive relations with others, environmental mastery, and autonomy [35]. Specific to the gig economy, Berger et al. surveyed Uber drivers in London and found that these drivers are more anxious than average London employees due to self-employment and instability. However, because of their love for gig work, a higher level of subjective well-being was observed. As a result of the platform's algorithms, drivers also often need to change their egos and spaces and upgrade their vehicles for financial incentives; these changes can damage the happiness they derive from their low-stress environment [36]. Other studies have found that gig jobs lead to overworking, lack of sleep, and fatigue. Disadvantages such as the lack of traditional job benefits (e.g., health insurance, compensation for overtime work) and possible exposure to COVID-19 also adversely affect gig workers' subjective well-being [37]. The researchers also found that PW contributes to organizational development [34]. Gig workers who suffer from uncertainty and lower income tend to have lower PW levels [38].

2.6. Paradoxical Leadership and Gigs

(1) PL emphasizes control and empowerment. It can simultaneously emphasize autonomy and control over decision making. Specifically, PL can control the behavior and roles of employees at work [30]. Simultaneously, employees are given flexible and autonomous freedom of movement. Empowerment benefits all employees by increasing autonomy and flexibility, as well as enabling employers to actively participate in solving problems. In this case, gig workers exhibit a discrete mindset that facilitates creative problem solving [33]. In addition to these practical benefits, self-determinism establishes autonomy as one of the basic needs of employees. Moreover, people are often trusted with solving problems creatively.

(2) PL can maintain distance and proximity. These two extremes allow employees to gain respect for their leaders while maintaining close relationships. Especially in the face of

gigs, which do not involve social contact, respect for social distance makes workers feel the warmth of society and enterprises [30]. Under PL, leaders not only maintain power, indicating the existence of a class structure, but also build social relations that emphasize close harmony. In this harmonious relationship, gig workers can express their opinions smoothly, support their leaders' views, and improve work performance and happiness.

(3) Paradoxical leaders treat subordinates in the same way while allowing personalization, which enables them to focus on the interests and skills of employees when assigning tasks [30]. Under such an arrangement, foresters receive assignments that are consistent with their abilities and interests and gain access to opportunities to develop their interests and skills.

Researchers have conducted in-depth studies on corporate performance [39]. The performance of CBEC companies is mainly divided into strategic performance and financial performance, with the former being a subjective measure of the realization of export objectives and with the latter being the objective indicator of financial results [16]. Previous studies have indicated that PL positively and significantly impacts both types of CP [31,32]. Paradoxical leaders can embrace complex environments [36], learn to think in opposition, encourage innovation, and deal with changes [37]. Harmonious internal team dynamics and better CP in turn encourage employees to be more skilled [40], adaptable, and proactive [30]. Therefore, PL can effectively improve organizational performance. Considering the above argument, we believe that gig workers serve as pillars in the development of CBEC in China. Moreover, psychological well-being greatly affects CP. In particular, gig workers with PL show significantly improved PW [41]. Thus, the relationship between PL, PW, and CP is described in the following assumptions:

Hypothesis H1 (H1): *The PL of CBEC gig workers has a positive influence on CP.*

Along this line of thought, we further believe that complex work environments and highly uncertain job opportunities prompt Chinese CBEC gig workers to perceive stagnation and misery in their career development. Prior research has found that paradoxical leaders play a positive role in influencing employees' concept selection, initiative, and work commitment [42,43]. Such leaders encourage employees to embrace and accept contradictions, help them make practical transformations and overcome difficulties through learning, and provide them with additional support and guidance, thereby enhancing their employees' pride and happiness in completing their work and being able to face challenges more positively. In this study, we infer that PL will drive Chinese CBEC gig workers' leadership identity on the basis of social identity theory. The encountered deficiencies in procurement and supply, logistics distribution, electronic payments, and after-sales service can challenge gig workers while guiding them to accept and embrace complex environments [44]. Therefore, we contend that gig workers may obtain flexibility, autonomy, and decision-making rights from their leaders. If they are led and encouraged by paradoxical leaders, they will increase their job satisfaction [45]. Therefore, we assume the following:

Hypothesis H2 (H2): *The PL of CBEC gig workers has a positive influence on PW.*

Scholars have found that PW is associated with several outcomes. For example, Cartwright and Cooper [46] argued that people with higher PW levels are happier, live longer, and are more productive. Burch and Cooper Brown [47] explained that PW leads to more self-stimulating performance than work ability and attitude. Research has proved that employees' low happiness will result in a large loss in organizational performance. [48] Happy employees are better able to serve customers, which leads to higher performance for both individuals and organizations. As PW is considered a positive proxy for burnout and well-being and there is a specific relationship between PW and positive performance outcomes at the individual and organizational levels [49], we formulate the following hypothesis that emphasizes an association between PW and CP:

Hypothesis H3 (H3): *The PW of CBEC gig workers has a positive influence on CP.*

Mental health (enjoyment and happiness) affects commitment to job performance through employee emotions. Commitment is also associated with employee happiness and personal growth [50]. Thus, higher levels of happiness are associated with higher levels of emotion among employees, which, in turn, improves job performance. In social exchange theory, behavior is the result of the exchange process [51]. These exchanges can be physical, social, or emotional. Employees can be more committed to the development of their organization if they perceive that their organization provides them with job security, takes care of their well-being, and gives them enough freedom to work [39]. Extending all the hypotheses above, it is reasonable to assume that the PW of Chinese CBEC gig workers may also act as a critical catalyst for transforming PL into CP [52]. Figure 1 presents the comprehensive framework of this research. Considering the theories cited in the figure, we propose the following hypothesis:

Hypothesis H4 (H4): *The PW of Chinese CBEC gig workers positively mediates the relationship between PL and CP.*

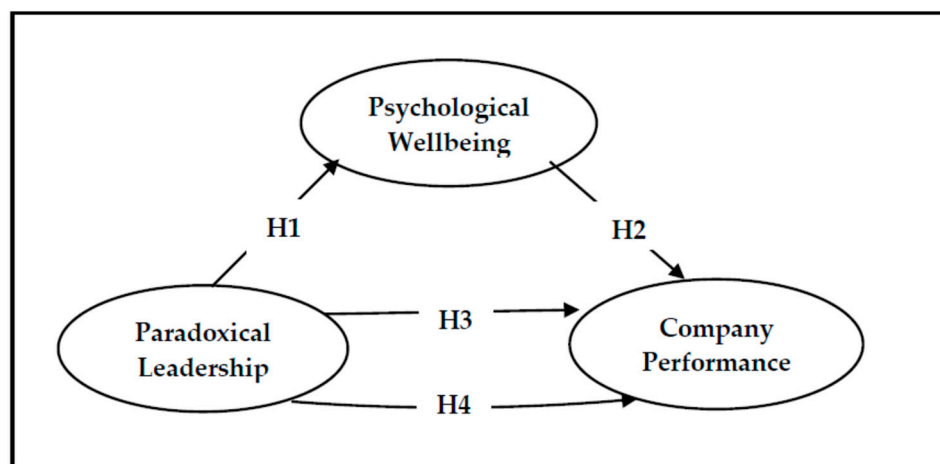


Figure 1. Research framework.

3. Methods

3.1. Data Collection

We selected CBEC enterprises in Zhejiang, Jiangsu, Jiangxi, and other provinces in China. Only gig workers were invited to answer the questionnaire to control for external influences. Before the formal investigation in March 2022, the researchers interviewed five local CBEC companies and conducted pilot tests with 70 employees. This ensured the accuracy and clarity of the questionnaire.

The questionnaires in this study were distributed on WJX (<https://www.wjx.cn/> accessed on 7 May 2021), a professional online survey website in China. The survey focused on Chinese CBEC gig workers' perceptions of the impact of PL and PW on CP. We adopted convenience and snowball sampling to expand the sample size of the questionnaire through sharing. We invited CBEC gig workers to share the questionnaire with their local gig community and encouraged gig workers to participate by giving them an incentive payment of RMB 2 each.

Considering the unstable working hours of CBEC gig workers, we designed the beginning of the questionnaire to be relatively simple to help them review their psychological states. A total of 346 questionnaires were collected in seven provinces through WJX. After excluding 31 invalid questionnaires, 316 valid questionnaires were collected. Of the respondents, 48.1% were female, and 51.9% were male; 68.04% were aged under 40 years; 31% were married, and 69% were unmarried; 57.59% had college degrees or lower; and 39.87%

had worked for their respective companies for less than one year. Moreover, 19.3% of the respondents were in production positions while 27.84%, 18%, 12.34%, 9%, 10.76%, and 2.2% were in technical positions, research and development positions, marketing positions, administrative (logistics, finance, personnel) positions, translation positions, and other positions, respectively. In addition, 24.68% of the respondents were from state-owned enterprises while the rest were from private enterprises.

3.2. Measures

The SmartPLS package was used to analyze the data [51]. Structural equation models were adopted to test the hypothetical mediation model. The participants were asked to answer on a six-point Likert scale (1 = strongly disagree, 6 = strongly agree) as it was confirmed to be valid.

PL was measured using five items according to Benach et al. [35]. The sample items included treating subordinates uniformly while allowing individualization (Cronbach's $\alpha = 0.938$). PW was measured [52] using seven items. The sample items included "I lead a purposeful and meaningful life" (Cronbach's $\alpha = 0.961$). CP was measured with four items according to Rodriguez et al. [39]. A sample item is "The number of employees has increased significantly" (Cronbach's $\alpha = 0.912$).

4. Results

4.1. Reliability and Validity

Cronbach's alpha and compound reliability (CR) values were used to test reliability. As shown in Table 1, the Cronbach's α and CR values for each dimension were in the range of 0.912–0.98 (PL) and 0.939–0.968, respectively. In addition, the new indicator coefficient roh_A introduced in SmartPLS 3.0 was evaluated to correct the estimates of the measurement structure; it ranged from 0.913 to 0.961 (PL) and was thus higher than the recommended threshold of 0.7, indicating that the internal consistency of our measurements was at a level that could be met [53].

Table 1. Confirmatory factor analysis.

Construct	Items	Factor Loading	α	CR	roh_A	AVE
PL	PL1	0.805	0.938	0.953	0.939	0.803
	PL2	0.915				
	PL3	0.932				
	PL4	0.91				
	PL5	0.913				
PW	PW1	0.789	0.961	0.968	0.961	0.812
	PW2	0.913				
	PW3	0.915				
	PW4	0.92				
	PW5	0.931				
	PW6	0.915				
	PW7	0.916				
CP	CP1	0.81	0.912	0.939	0.913	0.794
	CP2	0.917				
	CP3	0.926				
	CP4	0.905				

PL: paradoxical leadership, PW: psychological well-being, CP: company performance.

Convergence validity can be tested by checking whether the factor loading of all items and the mean average variance extracted (AVE) values are greater than 0.6 [54]. As shown in Table 1, the ranges of factor loadings and AVE are 0.789–0.931 (PL) and 0.794–0.812 (PW), respectively. Therefore, our measurement model had an acceptable convergent validity. As the value of the variance inflation factor (VIF) is less than 5, there is no multicollinearity problem.

For the assessment of discriminant validity, we referred to the threshold proposed by Fornell and Larcker [54]. The statistical results showed that the square root of each AVE value was greater than the correlation coefficient. The validity of the proposed method was thus verified. The results are presented in Table 2.

Table 2. Analysis of discriminant validity (Fornell–Larcker criterion).

Construct	Psychological Well-Being	Paradoxical Leadership	Company Performance
PW	0.901		
PL	0.845	0.896	
CP	0.841	0.855	0.891

4.2. Structural Model and Result Analysis

To test the hypotheses, we used the bootstrap resampling method in SmartPLS in evaluating the partial least squares (PLS) results; the responses were resampled 5000 times [55]. The results are presented in Table 3. The coefficient of determination, R^2 , of PL and PW relative to CP was 0.779, indicating that the PL and PW research models accounted for 78% of the variance in CP. In addition, the R^2 value of PL to PW was 0.712, indicating that the PL research model accounted for 71% of the variance in PW. Both had good explanatory power [53].

Table 3. Serial mediation results.

Hypothesis	Effect	T-Value	p-Value	Result
H1: PL $\rightarrow$ CP	0.855	41.97	0.0000	Significant
H2: PL $\rightarrow$ PW	0.845	38.001	0.0000	Significant
H2: PW $\rightarrow$ CP	0.413	4.948	0.0000	Significant
H4: PL $\rightarrow$ PW $\rightarrow$ CP	0.348	4.92	0.0000	Significant

The empirical findings supported Hypotheses H1, H2, H3, and H4. The results suggest that PL is positively related to CP (H1, $\beta = 0.41$ ***, $p < 0.001$), PL is significantly related to PW (H2, $\beta = 0.85$ ***, $p < 0.001$), and PW is significantly related to CP (H3, $\beta = 0.51$ ***, $p < 0.001$). The mediating assumption that PW mediates the relationship between PL and CP was verified (H4, $\beta = 0.35$ ***, $p < 0.001$). Figure 2 shows the PLS results for the research model.

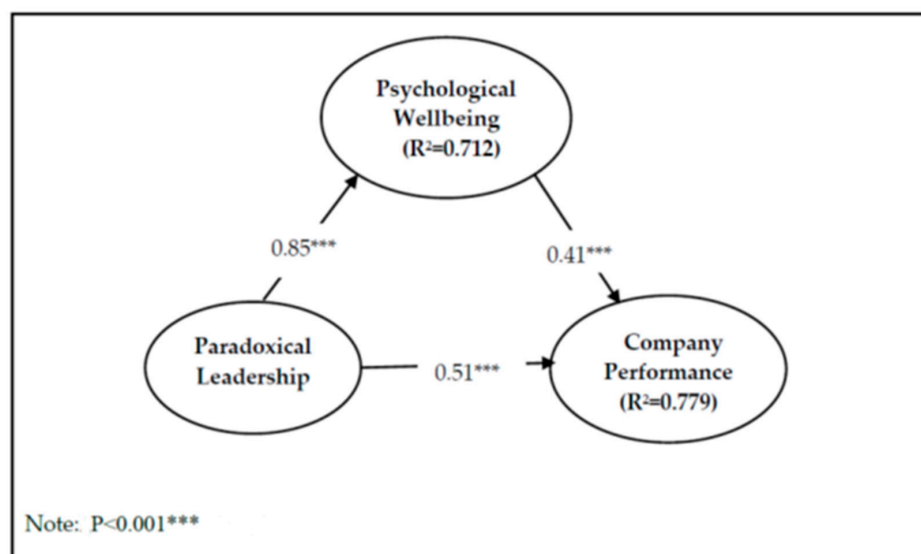


Figure 2. PLS results.

5. Discussion

5.1. Main Findings

Previous scholars have studied the relationship between PL and CP. However, these studies have mainly focused on the manufacturing industry. Hence, the current study is the first to investigate the role of PL in the CP of Chinese CBEC enterprises. It is also the first to focus on CBEC gig jobs. Our findings support the link between PL, PW, and CP. First, the results show that PL significantly affects the indirect relationships of CP; hence, H1 is supported. She et al. [56] conducted a large-scale survey of employees in the hospitality industry in China, and their findings support our results.

Second, PL emphasizes autonomy and decision control [55]. Autonomy enables employees to have more freedom to express themselves [34] while providing distance and intimacy. Paradoxical leaders treat subordinates uniformly but allow personalization. This allows them to make assignments with consideration of employees' interests and skills. PL keeps employees mentally and physically active, helps them feel energetic, and lifts their mood. Therefore, compared with traditional leadership, PL can promote the PW of employees. Our results also confirm that PL is positively and significantly correlated with PW; hence, H2 is supported. Zhang et al. [30] conducted a study on six private technology and engineering service companies in China and reported that PL positively affects PW.

Third, the findings of this study confirm a significant positive correlation between PW and CP, thus providing support for H3. Scholars conducted a survey of employees in the largest and most competitive private telecoms group in Pakistan [55], and their results support the outcomes of our study.

Finally, the outcome of this study verifies the mediating effect of PW on the relationship between PL and CP [17,57]. Employees who are not supported and recognized at work tend to give up, thus damaging company reputation and leading to customer dissatisfaction and reduced CP [22]. Therefore, the results of this study confirm that PW [16] positively mediates PL and CP, thus providing support for H4.

Overall, our findings support the four hypotheses. Some CBEC gig workers are under great stress because of instability. The stress comes from the nature of the gig work itself, unstable income, and great personal financial risk. The results of our study show that the PL of CBEC gig workers in China is positively correlated with CP and that PW plays a mediating role in the relationship between PL and CP [22]. More specifically, PL can effectively [4] improve the performance of the entire organization by increasing the subjective well-being of Chinese CBEC gig workers.

5.2. Contributions

We contribute to the existing literature in three aspects.

(1) Although the concept of PL only emerged in the 1980s, it has generally been adopted to solve strategic problems at the macro level. Our research provides a new direction and a new leadership behavior model for solving the organizational paradox of leadership. We advance this theory by demonstrating that PL should also focus on the mental health needs of individuals at the micro level. In other words, leaders face inherent paradoxes, structural demands, and challenges related to the mental health of their employees.

(2) Our research focuses on the increasing complexity of organizations and the business environment. Some management styles in the last century, such as transactional leadership and excellence leadership, may be effective in relatively stable and simple environments. However, in complex and dynamic environments, new leadership paradigms are needed, especially when leading groups of gig workers. Structures and needs are often at odds with employee needs. Requirements are relatively simple and easy to manage when they are linear and static. However, when the environment changes significantly, PL may be able to position itself more effectively in response to the increasingly complex conditions. Overall, finding the appropriate management approach amid environmental complexity has long been a challenge. In conclusion, this study broadens our perspective of leadership

styles in complex environments. Generally, PL has the ability to blur the boundaries of contradictions and encourage holistic thinking; broadening the cognitive base of “both” would be beneficial.

(3) Considering the reality of gig work in CBEC enterprises in China, this study proposes a new idea that PL can directly and effectively promote the performance of gig workers by improving their PW. Few studies have focused on the mental health and PW of CBEC enterprise and gig workers, and few scholars have conducted relevant empirical studies.

5.3. Management Implications

The results of this study also have certain guiding significance for the practice of CBEC enterprises.

(1) The results should help CBEC enterprises, gig workers, and their management to fully understand the importance of PL in the operations management in the context of CBEC. First, this study presents the definition and characteristics of CBEC enterprises. It also summarizes the connotations and concepts of PL, the characteristics of gig jobs in CBEC enterprises, and the main difficulties (e.g., job security, lack of work enthusiasm, and welfare deprivation). The outcomes will help Chinese CBEC companies manage gig jobs better.

(2) Enterprises should recognize and value the important strategic role of PL. Previous studies have shown that PL management concepts can effectively improve organizational performance and create value. Therefore, at the practical level, Chinese CBEC enterprises should acknowledge the important application of PL in managing gigs. Doing so will promote gig workers’ happiness and motivate them to complete work on time while meeting the highest quality and quantity standards. This improvement will in turn boost enterprise performance and competitive advantage.

(3) CBEC enterprises rely heavily on employee performance to accomplish organizational marketing. Following previous studies, we introduce the concept of PL influencing gig workers’ well-being and organizational performance to investigate the significant improvement in the performance of companies that mainly depend on employee performance after adopting different management methods. The results show that CBEC organizations need to pay more attention to and adopt appropriate management methods in the face of an increasingly changing environment. In this way, gig workers’ happiness can be promoted more effectively, difficulties can be faced, deficiencies can be overcome, and performance can be improved through joint consultation. Regardless of changes in the external environment, CBEC enterprises need to cooperate to effectively reduce business risks.

In the operation and management of CBEC enterprises, the management mode should be in line with organizational requirements and the actual implementation of gig work so as to maximize organizational benefits.

6. Conclusions

Owing to their low-paying and temporary work, Chinese CBEC gig workers do not typically maintain employee–employer relationships or receive standard work benefits. They also have no access to protection under labor policies, health insurance, skills training, career development, and retirement schemes; they also suffer from overprotection of consumers and inadequate savings [49,58]. Organizations and environments are becoming increasingly complex. The model of excellent leadership and management that has worked well in relatively stable environments in the last century is no longer suitable for complex and dynamic environments. CBEC needs a new leadership paradigm to address the structural nature of the market and the shifting demands of gig workers. Simple, linear, and static management methods can no longer be adopted. Meanwhile, PL can position itself more effectively in complex environments and develop ways to live dynamically amid complexities. CBEC enterprises can outsource work to small-scale microtasks, such

as research and development and customer service. Such flexible work content appeals to gig workers [31] who are not looking for full-time jobs but want to arrange their time freely.

This study proposes some useful suggestions from the perspective of sustainable CBEC development in China [58]. Sustainable development means finding a balance between three main factors: Sustainable development of the environment focuses on emphasizing the quality of the environment necessary to sustain economic activities and the quality of people's lives. Sustainable social development should ensure equal human rights, preserve culture and respect cultural diversity, race and religion. Sustainable economic development emphasizes the importance of maintaining a balance between people's natural social and human capital and their income standard of living [55]. There are three main ideas that people use to ensure the implementation of sustainable development. (1) Ensuring development, ecological constraints in line with the needs of social and economic development; (2) ensuring demand, redistributing resources on the basis of ensuring the quality of life for all; (3) ensuring future generations. Plan the use of resources in a way that ensures the necessary quality of life for future generations. [59,60] This study provides a possible way for China's CBEC to achieve equality in meeting the basic needs of gig workers, ensuring social awareness and cultural diversity. Specifically, it facilitates the selection of leadership methods by those engaged in CBEC management. Leadership with a certain degree of flexibility and autonomy will be particularly supportive of gig workers' performance and output by improving their PW as gig workers may require more flexibility and autonomy than traditional permanent workers [30]. According to the survey, highly educated gig workers require more autonomy and tolerance. Therefore, if CBEC enterprises manage work in an orderly manner and provide employees with more freedom, they can maintain the stability of gig jobs.

Our findings also provide new perspectives on sharing economy practices. As the instability of the sharing economy exceeds that of the traditional economic model, gig workers are underprivileged [41]. If PL can be widely adopted in CBEC, employees can be better supported so that they can have greater autonomy, and their PW and commitment to the enterprise can be improved. This ultimately translates to organizational performance [22]. Especially for the management of CBEC gig workers, PL [59] can serve as an effective catalyst to improve the PW of gig workers in a dramatically changing market environment and ultimately improve CP.

7. Limitations and Future Research

As a new economic form, CBEC has not been studied comprehensively. Many CBEC enterprises have a life span of 1–3 years [9]. Our research achieves useful results, especially in the context of COVID-19, and provides theoretical support for the development of CBEC in China [13].

However, this study has some limitations. First, the sample size is relatively small. Owing to the inconvenience caused by the epidemic, the sample size of this study was insufficient. Future studies may expand the sample size and its validity. Second, the samples in this study are mainly from Chinese provinces where CBEC companies are concentrated; hence, samples from central and western provinces are insufficient. Owing to gig workers' low job commitment [61], the results of this study may not be generalizable to other workers or industries. Future research should be carried out in more jobs and industries or in other countries with more developed sharing economies. Finally, most measures of PL are based on a sample of long-term workers; thus, the validity of the scales used in our survey is limited. Attention should be paid to the development of gig workers' scales in the future [31].

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